3rd Normal Form

A relation is in 3rd Normal Form if it is in 2nd NF and No transitive dependencies exist. it mean that for 3rd NF a non-key attributes must not depend on any other non-key attributes. its is also called Non-transitive dependency.

Example:

Consider a Relation. where Relation Name is Sales and CUSNO is Primary Key.

SALES

CUSNO	Name	SALEMAN	REGION
Coo1	HEWAD	Rashid	East
Coo2	Allied	Shahid	South
Coo6	AWan	Majid	East
Coo3	Hajvairy	Zahid	West
Coo8	Punjab	abid	South
Coo4	Hitech	Sadiq	East

these functional dependencies exist in the relation ✓ CUSNO → NAME, SALEMAN

✓ SALESMAN → REGION

- Sales table in 2nd NF because

Primary key consist only one attribute. But region only depends on SALESMAN therefore it is transitive dependency which is violation of 3rd NF.

as result there are update anomalies in relation SALES. A new salesman assigned to north region can not be entered until a customer has been assigned to him.

if CUSNO Coo4 we loose the Information that SALEMAN Imran is assigned to the East region.

if SALESMAN Muneer is reassigned to the East region, serval rows must be changed to reflect the fact.

These annomlies arieses as result of the transitive dependency. This problem can be removed by decomposing the relation.

SALESMAN

CUSNO	NAME	SALEMAN
Coo1	HENAD	Rashid
Coo2	Allied	Shahid
Coo6	AWAN	Majid
Coo3	Hajvairy	Zahid
Coo8	Punjab	abid
Coo4	Hitech	Sadiq

Now both relation are in 3rd NF because no transitive dependency exist

SALES

SALESMAN	Region
Rashid	East
Shahid	South
Majid	East
Zahid	West
abid	South
Sadiq	East